

Retrieval That Improves the Longitudinal Patient History

Finding What Is Known — and What Still Needs to Be Known

Synopsis

A national longitudinal patient history will contain information from many healthcare systems, payers, and patients. Some of that information will conflict, and some treatments will have no recorded outcome. AI should reconcile differences when the evidence is sufficient, but it should not manufacture certainty or create another documentation burden for providers.

Retrieval provides an opportunity to improve the history as it is used. Clinically relevant ambiguities can be surfaced for resolution, while missing treatment outcomes can be collected from patients or providers and associated with the original treatment. Over time, this can improve clinical knowledge while connecting treatments and costs with their results.

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Introduction:

Most discussions of retrieving a longitudinal patient history focus on finding the information relevant to the current clinical decision. But retrieval can reveal something equally important: **the history itself may contain questions that have never been resolved.**

Two types are particularly important. Information assembled from different healthcare systems, payers, and patients may conflict in ways that AI cannot reliably reconcile. And the history may document a treatment in considerable detail without ever recording whether it worked.

AI should not hide either problem. Uncertain information should remain uncertain until sufficient evidence resolves it, and missing outcomes should be recognized as missing rather than inferred from the absence of subsequent treatment.

Retrieval creates an opportunity to address both problems when they become relevant and when the patient or provider can supply the missing knowledge. Done selectively, this can improve the longitudinal history without creating another documentation burden for providers.

Retrieve → Identify Unresolved Knowledge → Ask → Resolve → Update → Reuse

When the Sources Disagree

Information assembled from multiple healthcare systems will inevitably contain conflicts. A hospital may have one version of a medication history while the PCP has another. A specialist may identify a condition as active while another record indicates that it was resolved. The patient may provide information that differs from the clinical record.

Many differences can be reconciled automatically. AI may determine that differently worded diagnoses represent the same condition, that a newer medication order superseded an older one, or that a procedure documented by a provider corresponds to a payer claim.

Other conflicts cannot be resolved reliably from the available evidence. In those cases, requiring AI to select the most probable answer risks turning uncertainty into apparent fact.

Suppose a patient reports that a condition was diagnosed in 2021, a specialist record first documents it in 2023, and the PCP's problem list does not contain it. The longitudinal history should retain the conflicting evidence and its provenance rather than arbitrarily selecting one version.

When that information becomes relevant, retrieval could surface the issue to the provider:

Unresolved information: Conflicting evidence exists concerning this diagnosis and its onset.

The provider could review the evidence and, when appropriate, ask the patient. The resulting resolution would become part of the longitudinal history, together with who resolved it, when it was resolved, and the evidence supporting the decision.

The same principle applies when the patient can resolve a question directly. Patients may know whether they actually took a medication, stopped taking it, received a procedure, saw a referred specialist, or experienced improvement after treatment. Patient-provided information should retain its provenance and should not automatically override clinical evidence, but the patient is an important source for resolving parts of the longitudinal history.

The objective is not to eliminate uncertainty artificially. It is to preserve uncertainty until sufficient evidence exists to resolve it.

Ambiguity in, ambiguity out. Resolve once. Correct when necessary. Use many times.

Do Not Turn Ambiguity into Provider Work

When the national longitudinal patient history is first assembled, it will expose an enormous backlog of conflicting, incomplete, and ambiguous information accumulated over many years and across many healthcare organizations. Attempting to resolve that backlog by presenting discrepancies to physicians would be impractical and counterproductive.

Healthcare technology already risks increasing provider workload when obtaining the benefit of the technology requires additional data entry. A longitudinal architecture should be designed around the opposite principle: **resolution should reduce future work, not create a new documentation burden.**

AI should therefore resolve differences automatically when the evidence is sufficient and retain low-value uncertainty without requiring anyone to address it. Historical ambiguities need not be resolved merely because they exist. Most can remain dormant unless they become relevant to a future clinical decision.

When human input is necessary, the work should be directed to the appropriate person. Some questions require clinical judgment. Others—whether a patient took a medication, saw a referred specialist, or improved after treatment—may be answerable by the patient.

Provider attention should be reserved for unresolved information that is clinically important to the current encounter and cannot be resolved reliably from the available evidence.

Even then, resolution should be incorporated into the normal clinical workflow rather than added as another documentation task. If a physician asks a patient whether a previous treatment worked and the answer is captured in the encounter conversation or note, AI should be able to associate that information with the earlier treatment and update the longitudinal history with appropriate provenance. The objective should be fewer clicks and forms, not more.

This also changes how we should view the initial backlog. The national longitudinal history does not need to become perfectly reconciled before it becomes useful. Historical ambiguity can be resolved incrementally as patients interact with healthcare and as particular questions become relevant. Each resolution then becomes available to authorized future users.

The governing principle should be:

Do not ask a human what the evidence can resolve. Do not resolve what does not matter. Do not ask a provider what the patient can answer. And do not ask twice for information that has already been responsibly resolved.

When Healthcare Records the Treatment but Not the Outcome

A different problem occurs when there is no conflicting information because the system never captured it.

Healthcare systems are effective at recording activity: a problem was identified, a referral was made, a specialist was seen, a treatment was provided, a prescription was written, and a claim was paid. They are much less consistent about recording whether the treatment actually helped the patient.

This is particularly likely when a specialist provides treatment, and there is no follow-up appointment.

A personal experience illustrates the problem. Hip bursitis had reached the point where I could walk only about one-quarter mile before needing to stop and rest. My primary care physician referred me to an orthopedic specialist, who treated the condition with cortisone injections and prescribed pain medication with three refills.

The treatment worked. I gradually increased the distance of my walks and eventually no longer needed the pain medication, so I canceled the final refill. There was no follow-up

appointment with the orthopedic physician, and at subsequent visits my PCP never asked what happened with my hip.

The healthcare system therefore recorded the problem, referral, specialist encounter, treatment, prescriptions, refills, and associated claims. It probably never recorded the most important information: **the treatment substantially improved my ability to walk and eliminated my need for the pain medication.**

Research has documented longstanding problems in “closing the referral loop” between primary care and specialists, including incomplete referrals and failures to return specialist information to primary care. But completing the referral and returning the specialist's report still does not necessarily close the clinical loop. A specialist may assess and treat the patient, document the treatment, and never see the patient again. The referring physician may receive the specialist's report but never subsequently ask whether the treatment worked. The referral has been closed; the **treatment-outcome loop has not.**¹

An AI system could infer something from the absence of additional orthopedic treatment or from cancellation of the final refill. But those are signals, not outcomes. The same utilization pattern could mean the treatment worked, the patient decided to tolerate the pain, the medication caused side effects, the patient sought treatment elsewhere, or the patient simply did not return.

Absence of additional treatment is not evidence of a successful outcome.

Instead of making that inference, retrieval should recognize that a treatment occurred and that no subsequent outcome was recorded. At my next PCP visit, the relevant history could include a simple prompt:

Outcome unresolved — hip bursitis: Orthopedic treatment included cortisone injections and pain medication. No subsequent assessment of pain or walking limitation has been recorded.

The PCP could ask one question: “How did your hip respond to the treatment?”

¹ Malhar P. Patel et al., “Closing the Referral Loop: an Analysis of Primary Care Referrals to Specialists in a Large Health System,” *Journal of General Internal Medicine*, 2018. The study analyzed 103,737 primary-care referral scheduling attempts across 20 specialties; only 34.8% resulted in documented completed specialist appointments. The authors define “closing the referral loop” as completion of the specialist appointment with results available to the referring PCP. [PubMed](#)
[PubMed: PMID 29532299](#)

My answer could then be recorded as a patient-reported outcome associated with the earlier orthopedic treatment: walking tolerance increased substantially and pain medication was no longer required.

The important architectural point is that the PCP did not provide the treatment. A longitudinal architecture should nevertheless allow the PCP to record its outcome.

Outcomes belong to the longitudinal patient history, not exclusively to the healthcare information system of the provider who delivered the treatment.

The Missing Outcome Is Also Missing Economic Information

The same missing outcome also affects the payer.

Suppose the orthopedic treatment had been substantially more expensive and required prior authorization. Before approving it, the payer might receive extensive information about the diagnosis, previous treatments, proposed procedure, medical necessity, provider, and expected cost. Afterward, the payer would receive the claim and know what it paid.

It might never learn whether the treatment worked.

The information sequence would effectively be:

Condition → Authorization → Treatment → Cost → ?

Claims can show what services were delivered and what they cost. Subsequent claims can provide indirect evidence about what happened afterward. But utilization is not an outcome. A patient who receives no additional treatment may have recovered completely or may still have the same functional limitation.

Capturing outcomes would allow the longitudinal history to connect information that is usually separated:

Condition + Severity → Treatment → Cost → Outcome

Across many patients, those relationships could provide a foundation for comparing risk-adjusted costs and outcomes. Providers and payers could better understand which treatments work for which patients, whether benefits persist, when additional treatment is required, and whether more expensive interventions produce correspondingly better outcomes.

This does not require turning every clinical encounter into a research study. Often the missing information may be simple: Did the treatment help? Is the pain gone? Can the patient walk farther? Is the medication still needed?

The longitudinal history can identify when an important outcome is missing and retrieve that question when someone who can answer it is already interacting with the healthcare system.

Retrieval as a Feedback Mechanism

These two problems—conflicting information and missing outcomes—appear different, but architecturally they represent the same condition: **the longitudinal history contains unresolved knowledge.**

In one case, multiple sources provide evidence, but the evidence cannot yet be reconciled. In the other, the record contains an intervention but lacks the information needed to understand its result.

Neither should be hidden by AI.

Instead, the longitudinal architecture should explicitly represent unresolved knowledge, along with its supporting evidence and provenance. AI can determine whether the unresolved issue is relevant to the current encounter and whether the provider or patient can reasonably resolve it.

That relevance test is essential. A longitudinal history could contain many uncertainties and missing outcomes. Presenting all of them would simply create another form of information overload. The objective is to surface unresolved information when it matters to the current clinical decision or when the encounter provides a reasonable opportunity to obtain information of continuing value.

Resolution should also preserve provenance. A patient's report that a treatment worked is not the same as a physician's clinical assessment or a laboratory measurement. Each can be useful if the source, date, related treatment, and basis of the information remain visible.

Retrieval therefore becomes part of a feedback loop:

Retrieve → Identify → Ask → Resolve → Update → Reuse

The longitudinal history does not merely accumulate additional records. Each appropriate use of the history becomes an opportunity to make it more complete and reliable.

From Historical Records to Longitudinal Knowledge

Healthcare information systems are necessarily designed around the organizations they serve. They record the organization's interactions with a patient—encounters, diagnoses, orders, tests, treatments, prescriptions, referrals, billing, and other information needed to

deliver and administer care. Each system therefore contains an important but incomplete view of the patient's healthcare.

A national longitudinal patient history starts from a different perspective: **the patient**. Provider and payer systems contribute information from their interactions with that patient to create knowledge that spans organizations and time. The objective is not to replace those systems, but to use what each knows to create a more complete history for the patient and those responsible for the patient's care.

That patient-centered perspective can reveal what no individual organization is positioned to see. A specialist may know what treatment was provided but never see the patient again. A PCP may see the patient later without realizing the specialist's treatment outcome was never recorded. A payer may know what the treatment cost without knowing whether it improved the patient's condition. The patient may know the answer, but no system asks the question.

The longitudinal history can connect those pieces. Its purpose therefore extends beyond asking, **“What do we know about this patient?”** It can also ask, **“What important information do we still need to know?”**

AI should reconcile information when the evidence supports doing so. When it does not, ambiguity should remain visible until it can be responsibly resolved. When a treatment has no known outcome, the absence should become a question, not an assumption. Each appropriate resolution or newly collected outcome improves the history for future care and, collectively, our understanding of healthcare effectiveness.

Resolve once. Correct or collect missing outcomes when necessary. Use many times.